Unitree G1 policy evaluation

Reproduce behavior. Measure the result.

Three deterministic 30-second MuJoCo runs at different forward commands, using the official Unitree G1 pretrained policy. Prepared by Ley.

Attribution: Unitree provides the pretrained policy and robot assets. I built the headless measurement adapter and this report. I did not train the policy. These results do not establish Go2 or physical-robot performance.

Measured tracking

Command (m/s)	Mean forward (m/s)	Forward RMSE (m/s)	Lateral RMS (m/s)	Duration (s)
0.0	0.014	0.017	0.077	30
0.2	0.201	0.017	0.076	30
0.5	0.477	0.034	0.065	30

Body-frame velocity. Metrics use samples from t = 2 s onward. Each row is one run; all completed the selected horizon without triggering the chosen fall rule. Survival alone is not task success.

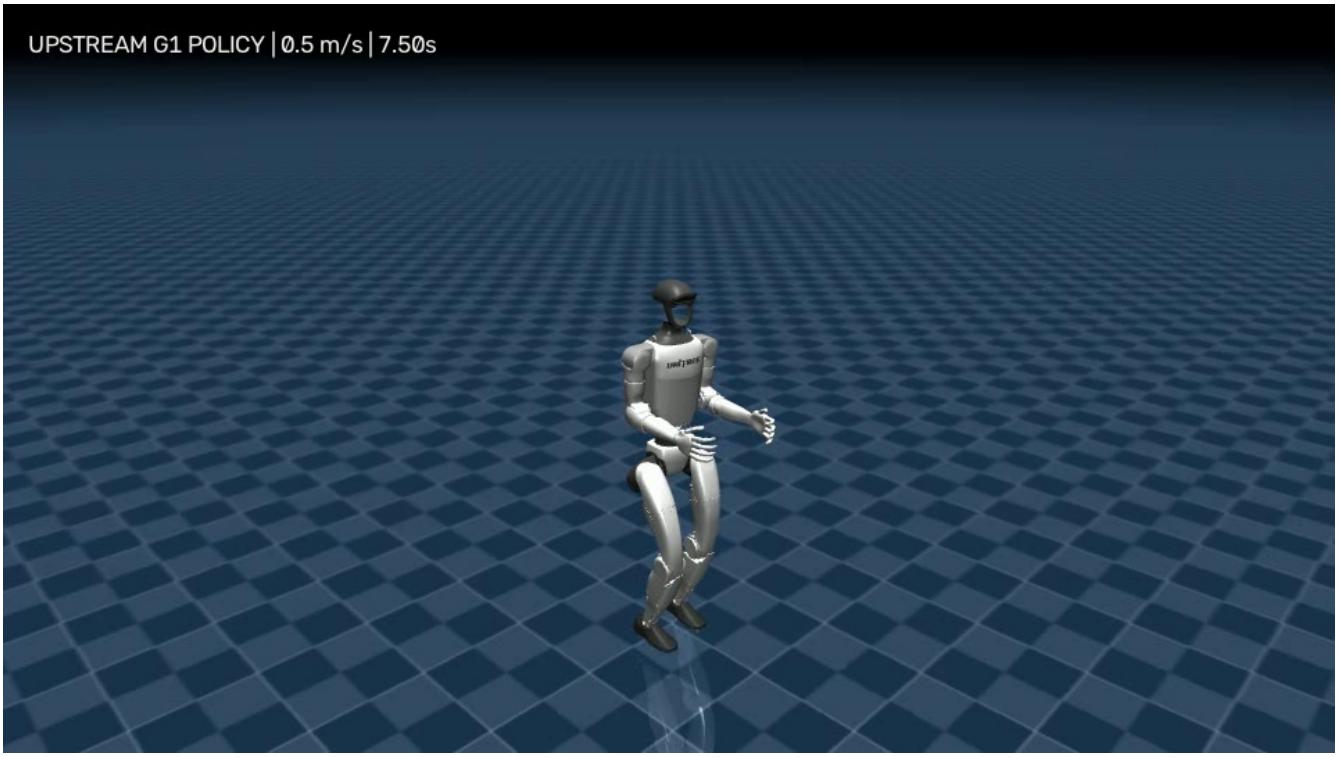
A useful limitation appears at zero command

At a zero forward command, mean forward speed was small, but lateral RMS was 0.077 m/s and final horizontal displacement from the origin was approximately 0.71 m. A standing-still requirement needs its own tolerance. This sample does not identify the cause of that lateral motion.

Engineering contribution

Added a fixed simulation-time horizon, command selection, body-frame velocity measurements, fall flags, file hashes and continuous video capture. Preserved the upstream observation, PD control and policy-update sequence.

A result that can be checked



Frame from a separate 15-second simulation at 0.5 m/s, recorded continuously at 50 fps. The table uses the longer 30-second measurement runs. Video available as a separate portfolio attachment.

Reproduction record

MuJoCo 3.11.0; PyTorch 2.13.0+cpu; CPU inference.
Physics step: 0.002 s. Policy update: 0.020 s.
One default initial state; no external perturbations.
Fall flag: base height below 0.4 m, upright cosine below 0.5, or nonfinite state.

Scope of evidence

This is a deployment-measurement example. It does not establish terrain robustness, push recovery, repeated-trial success rates, a new training method, hardware safety or sim-to-real transfer. No client source, weights or private logs are included.

Source and license

github.com/unitreerobotics/unitree_rl_gym
Commit: 276801e46c5d433564f24658bac64f254b7d2d4b
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